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Write legibly showing all the steps VERY clearly. (Total points = 110, **Max Score is =100.**)

Let $\Omega \neq \emptyset$, where $\emptyset$ = empty set and $\mathcal{P}(\Omega)$ is the collection of all subsets of Ω, i.e the power set. Let $\mathcal{G}, \mathcal{F} \subset \mathcal{P}(\Omega)$ be a classes of subsets of Ω, and λ, μ be set functions.

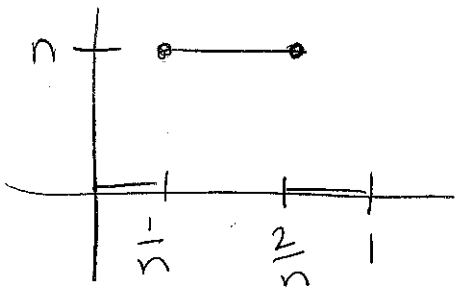
1. Let m be the Lebesgue measure on $[0,1]$ and $\mathbf{1}_A(x)$ denote the indicator function of A. Define a sequence of measurable transformations on $(\Omega = [0,1], \mathcal{F} = \mathcal{B}[0,1], \mu = m)$: For $n \geq 1$, and $x \in \Omega = [0,1]$,

$$f_n(x) = n\mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}(x) = \begin{cases} n & \text{if } \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & \text{otherwise} \end{cases}$$

The limit function is $f(x) \equiv 0$ for all $x \in \Omega = [0,1]$.

(10+10+10+10+10 = 50 points)

(a) Show that $f_n \to f$ almost everywhere (m).



Let $x \in (0,1]$

$\exists N_x \in \mathbb{N}$ s.t.

$x \notin [\frac{1}{n}, \frac{2}{n}]$

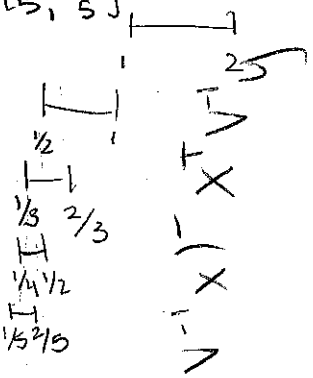
$\forall\, n \geq N_x$

$\therefore \lim_{n\to\infty} n \mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}(0)$

$\therefore \lim_{n\to\infty} \int n \mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}\, dm$

$= \lim_{n\to\infty} \int 0\, dm = 0 = f$

n=1 $[1, 2]$
n=2 $[\frac{1}{2}, 1]$
n=3 $[\frac{1}{3}, \frac{2}{3}]$
n=4 $[\frac{1}{4}, \frac{1}{2}]$
n=5 $[\frac{1}{5}, \frac{2}{5}]$



If $x = 0$, as $n \to \infty$, $\frac{1}{n} \to 0$ & $\frac{2}{n} \to 0$

$x = 0$

$\therefore \lim_{n\to\infty} \int n \mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}\, dm$

$[\frac{1}{n}, \frac{2}{n}] \to [0,0] = \{0\}$

$\therefore m(\{0\}) = 0 = f$

$\therefore \{x : \lim_{n\to\infty} \int n \mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}\, dm \neq \int 0\, dm\} = \emptyset$ X -5

$m(\emptyset) = 0 \therefore f_n \to f$ a.e. (m)

(b) Does the sequence $f_n \to f$ in measure?

$m([0,1]) = 1 < \infty$

& $f_n \to f$ a.e(m)

$\Rightarrow f_n \to f$ in measure

✓

(c) For what values of $0 < p < \infty$ does the sequence $f_n \to f$ in L_p?

$f_n = n I_{[\frac{1}{n}, \frac{2}{n}]}$ $\quad f \equiv 0$

$$\int |f_n - f|^p dm = \int n^p I_{[\frac{1}{n}, \frac{2}{n}]} dm = n^p \left(\frac{2}{n} - \frac{1}{n}\right) = n^p \left(\frac{1}{n}\right) = n^{p-1}$$

iff $0 < p < 1$ $\quad n^{p-1} \to 0$ as $n \to \infty$

$\therefore f_n \to f$ in L_p if $0 < p < 1$

& $f_n \not\to f$ in L_p if $p \geq 1$

$p = 1$ $\quad \int |f_n - f|^p dm = \int n I_{[\frac{1}{n}, \frac{2}{n}]} dm$

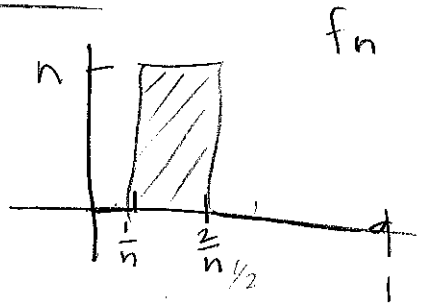
$= n^0 = 1 \not\to 0$ as $n \to \infty$

$\therefore f_n \not\to f$ in $L^1 \Rightarrow f_n \not\to f$ in L^p for $p > 1$

✓

(d) Is this sequence $\{f_n\}$ uniformly integrable?

if UI, $\lim_{t\to\infty}\left(\sup_{n\geq 1}\int_{\{|f_n|>t\}}|f_n|\,dm\right)=0$



(1) fix $t>0$

(if $n>t$) $\int_{\{|f_n|>t\}}|f_n|\,dm = \int_{\{|f_n|>t\}} nI_{[\frac{1}{n},\frac{2}{n}]}\,dm$

$= \int nI_{[\frac{1}{n},\frac{2}{n}]}\,dm = n\left(\frac{2}{n}-\frac{1}{n}\right)=1$

(2) $\sup_{n\geq 1}\int_{\{|f_n|>t\}}|f_n|\,dm \geq 1$

(3) $\lim_{t\to\infty} 1 \leq \lim_{t\to\infty}\left(\sup_{n\geq 1}\int_{\{|f_n|>t\}}|f_n|\,dm\right)$

$\Rightarrow 1 \leq \lim_{t\to\infty}\left(\sup_{n\geq 1}\int_{\{|f_n|>t\}}|f_n|\,dm\right)\neq 0$

$\therefore \{f_n\}$ is not UI ✓

$m \to$ Leb

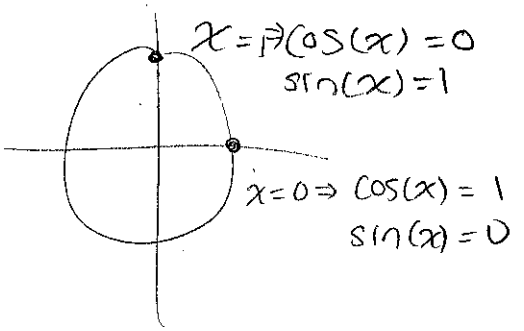
Name: $[0,1], \mathcal{B}([0,1])$ Stat 6410 (Ghosh) Midterm 2 **(Page 5 of 12)**

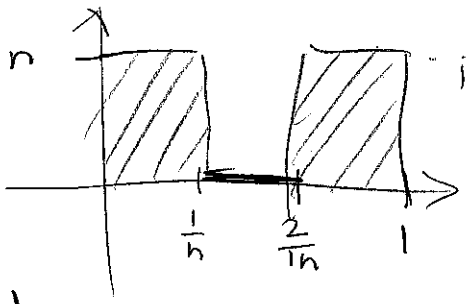
(e) Consider the sequence $\{g_n\}$ where $g_n = cos(f_n)$ (here $cos(x) = cosine(x)$ is the well-known trigonometric function). Does this sequence $\{g_n\}$ converge to $g \equiv 1$ almost surely (m)?

$f_n(\omega) = n 1_{[\frac{1}{n}, \frac{2}{n}]}(\omega)$

$\cos\left(n 1_{[\frac{1}{n}, \frac{2}{n}]}(\omega)\right) = \begin{cases} 0 & \omega \in [\frac{1}{n}, \frac{2}{n}] \\ n & \omega \notin [\frac{1}{n}, \frac{2}{n}] \end{cases}$

↑ can only be 1 or 0





intuitively, as $n \uparrow$ $[\frac{1}{n}, \frac{2}{n}]^c$ widens

$\{\omega : \lim_{n\to\infty} \int_{[0,1]} n 1_{[\frac{1}{n}, \frac{2}{n}]} dm \neq \int_{[0,1]} 1 \, dm$

$\int_{[0,1]} 1 \, dm = m([0,1]) 1$

$= 1$

$= \{\omega : \lim_{n\to\infty} (n-1) \neq 1\}$

$= [0,1]$

$\int n 1_{[0,\frac{1}{n}) \cup (\frac{2}{n}, 1]}(\omega) \, dm$

Since $m([0,1]) \neq 0$,

$= n \, m([0,\tfrac{1}{n}) \cup (\tfrac{2}{n}, 1]) dm$

$= n \left((\tfrac{1}{n} - 0) + (\tfrac{n}{n} - \tfrac{2}{n}) \right)$

$\therefore g_n \not\to g$ a.s(m)

$= n \left(\tfrac{1}{n} + \tfrac{n-2}{n} \right)$

$= n \left(\tfrac{n-1}{n} \right) = n - 1$

$\not\to 1$

2. (i) Show that a $Binomial(n = 10, p = 0.1)$ random variable exists:
In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \to \mathbb{R}$, such that its p.m.f. is:

$$P(X = k) = f(k) = \binom{10}{k}(0.1)^k(0.9)^{10-k},\ k \in \{0, 1, \ldots, 10\},$$

and $P(X = k) = f(k) = 0$, otherwise. $\{0,1,\ldots,10\} = \Omega$

(ii) Compute the following Lebesgue integrals (give a numerical answer)

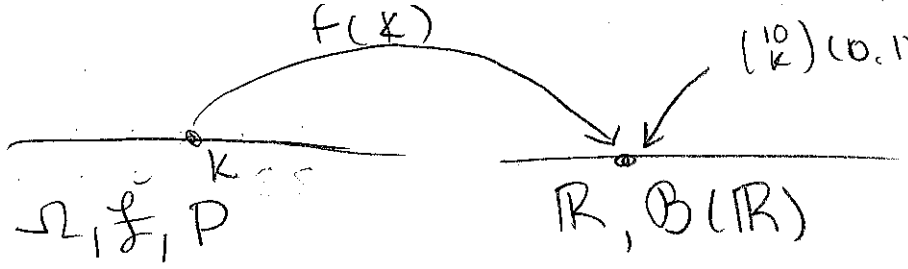
$$\int_\Omega X^2(\omega)dP(\omega)$$

I forgot to study this

You can state & use any pre-measure theory facts you know (if needed), but justify all the measure theory steps.

(10+15 = 25 points)

(i)



-5

$$F(k) = \sum_{K=0}^{k}\binom{10}{K}(0.1)^K(0.9)^{10-K} = P(X \le k)\ \forall\ k \in \{0,1,\ldots 10\}$$

F(k) is a cdf ∴ F is monotonically increasing & right continuous

$\mu_F([a,b] \in \mathcal{P}(\Omega)) = F(b) - F(a)$ ← Lebesgue Stil. measure WRT F

X is a binomial R.V

-8

$VAR(X) = EX^2 - (EX)^2$ $E(X) = n \times P = 10(0.1) = 1$

$\Omega = \{0, 1, 2, 3, \ldots, 10\}$

(ii) $\int_{\Omega} X^2(\omega)\, dP(\omega) = \int X^2(\omega)\, I(\omega \in \{0,1,\ldots,10\})\, dP(\omega)$

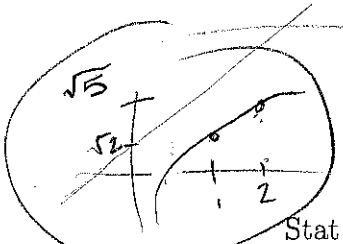
$= \sum_{\omega \in \{0,1,\ldots 10\}} P(X^2 = \omega) = \frac{19}{10}$

$\rightarrow EX^2 = Var$

$10\left(\frac{1}{10}\right)\left(\frac{9}{10}\right) = \frac{9}{10}$

$np(p-1) + np = \frac{9}{10} + 1 = \frac{19}{10}$

$var(X) + EX$



Name:

3. Answer the following:

(15+10+10 = 35 points)

(a) For a *nonnegative* random variable X, prove $\sqrt{1+(EX)^2} \le E\sqrt{1+X^2} \le 1+EX$, considering the possibility that $E(X)$ can be infinite.

If $X < \infty$ $g(x) = \sqrt{1+x}$ is clearly concave

$g''(x) = \frac{3}{2} 2x^2 (1+x^2)^{-5/2} - (1+x^2)^{-3/2} < 0 \quad \forall x \ge 0$

$E(g(X)) \le g(EX) \Rightarrow \sqrt{1+(EX)^2} \le E\sqrt{1+X^2}$ by Jensen's inequality

$\sqrt{1+x^2} \le 1+x \quad \forall x \ge 0$

$\therefore E(\sqrt{1+X^2}) \le E(1+X) = 1+EX$

combining ★ & ♡

$\sqrt{1+(EX)^2} \le E\sqrt{1+X^2} \le 1+EX$ ✓

If $EX = \infty$

$(EX)^2 = \infty \therefore \sqrt{1+(EX)^2} = \infty$ since $\sqrt{1+x^2}$ is ↑ fn

$E(\sqrt{1+X^2}) = \infty$

$1+EX = \infty$

$\sqrt{1+(EX)^2} \le E(\sqrt{1+X^2}) = 1+EX \Rightarrow$

$\sqrt{1+(EX)^2} \le E(\sqrt{1+X^2}) \le 1+EX$ ✓

(b) Let $(\Omega = \mathbb{R}, \mathcal{F} = \mathcal{B}(\mathbb{R}))$. Let m be the Lebesgue measure and μ be the standard normal (Gaussian) measure. That is,

$$\mu(A) = \int_A \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dm(x), \; A \in \mathcal{B}(\mathbb{R})$$

Show that the RN derivative of Lebesgue measure m w.r.t μ to be:

$$\left(\frac{dm}{d\mu}\right)(x) = \sqrt{2\pi} e^{\frac{x^2}{2}}, x \in \mathbb{R}.$$

(you can state any result proved in class to argue this)

We know from example 4.1.1 that $m \ll \mu$

$m(A) = \int_A \frac{dm}{d\mu} d\mu$ $\frac{dm}{d\mu}$ is the leb. measure

$\frac{dm}{d\mu} = \left(\frac{d\mu}{dm}\right)^{-1} = \frac{1}{\frac{1}{\sqrt{2\pi}} e^{-x^2/2}} = \sqrt{2\pi}\, e^{x^2/2}$ a.e.(μ) $\forall x \in \mathbb{R}$

✓

(c) Let $(\Omega, \mathcal{F})$ be a measurable space with two measures μ and ν on it. Given,

$$\mu \ll \nu, \text{ and } \mu \perp \nu.$$

Show that $\mu(A) = 0$ for all $A \in \mathcal{F}$.

from $\mu \perp \nu$, we know $\exists\ B \in \mathcal{B}(\mathbb{R})$
s.t. $\nu(B) = 0$ & $\mu(B^c) = 0$ [b]
[$B \cap B^c = \emptyset$ & $B \cup B^c = \Omega$ $\in \mathcal{B}(\mathbb{R})$]
∴ because μ is a measure
$\mu(B \cup B^c) = \mu(B) + \mu(B^c) = 0$
∴ $\forall\ A \in \mathcal{F}$, $A \subset \Omega$
$\Rightarrow \mu(A) = 0$ (since $\mu(\Omega) = 0$)
[monotonicity]

& b/c $\mu \ll \nu$, $\mu(B) = 0$ ✓